8 **(a)** Explain what is meant by

(i) an isotope.

.....

..... [1]

(ii) radioactive decay.

.....

.....

..... [2]

(iii) half-life of a radioactive substance.

.....

.....

..... [2]

(b) The number of atoms present in a radioactive source when the source was first acquired in 1990 was 2.5×10^{14}.

In 2018, the number of atoms that remains is 1.3×10^{14}. Determine the number of atoms that will remain in 2032.

number of atoms = ... [4]

(c) Strontium-90 $\left({}^{90}_{38}\text{Sr}\right)$ undergoes β-decay to form yttrium (Y).

Yttrium is itself radioactive and decays by the emission of a β-particle. The yttrium decays to form zirconium (Zr) which is stable.

Complete the nuclear equations below for the decay of strontium-90 to form zirconium.

$${}^{90}_{38}\text{Sr} \rightarrow \text{Y} +$$

$$\text{Y} \rightarrow \text{Zr} +$$

[2]

(d) Plutonium-239 decays by α-emission to form the isotope uranium-235. In this nuclear reaction, 5.26 MeV of energy is released. The energy of the emitted α-particle is 5.15 MeV.

Suggest and explain why these two values are different.

.....

.....

..... [2]

(e) Fig. 8.2 lists various nuclides and their respective mass and binding energy per nucleon.

Particle/ Nuclide	Name	Mass / u	Binding energy per nucleon / MeV
${}^{1}_{1}\text{H}$	Proton	1.00728	0
${}^{1}_{0}\text{n}$	Neutron	1.00866	0
${}^{4}_{2}\text{He}$	Helium	-	7.07470
${}^{14}_{7}\text{N}$	Nitrogen	-	7.47724
${}^{17}_{8}\text{O}$	Oxygen	-	7.75224

Fig. 8.2

(i) Explain why the binding energy of a proton or neutron is zero.

.....

..... [1]

(ii) Using the values from Fig. 8.2, determine the mass of helium nucleus, expressing your answer to an appropriate number of significant figures.

mass = ... u [3]

(iii) In a particular nuclear reaction, a slow moving alpha particle bombards a stationary nitrogen nucleus ${}^{14}_{7}N$, transmuting it to an oxygen nucleus ${}^{17}_{8}O$ and a proton.

The nuclear reaction is shown below.

$$ {}^{14}_{7}N + {}^{4}_{2}He \rightarrow {}^{17}_{8}O + {}^{1}_{1}H $$

Making use of values from Fig. 8.2, determine quantitatively whether the reaction can be spontaneous. Explain your workings.

[3]

End of Paper